Quantification: 543-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	12
interior lacuna count	12
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 12)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2027.92	2066	537.141	px^2
major_axis_length_px	lacuna shape	81.5016	86.4661	16.9465	px
minor_axis_length_px	lacuna shape	33.8846	32.3799	6.9212	px
aspect_ratio	lacuna shape	2.4727	2.3986	0.6666	unitless
eccentricity	lacuna shape	0.8942	0.9087	0.0558	unitless
solidity	lacuna shape	0.8887	0.9115	0.0668	unitless
roots_count	headline	6	6.5	3.1042	count
ring_length_r30_px	headline	246.083	241	110.019	px
ring_length_r60_px	reported	792.5	779.5	251.868	px
owned_length_px	ownership-dependent	2528.4	2419.86	1628.64	px
edge_count	ownership-dependent	109.25	105.5	67.9761	count
mean_edge_length_px	ownership-dependent	24.1146	23.1195	4.6014	px
perimeter_px	normalised	222.101	229.693	45.067	px
ring_area_r30_px2	normalised	8742.17	8905.5	1127.78	px^2
ring_area_r60_px2	normalised	22189.9	23189	2825.36	px^2
in_frame_fraction_r30	normalised	0.9965	1	0.0123	unitless
in_frame_fraction_r60	normalised	0.9814	1	0.0596	unitless
ring_density_r30	normalised	0.0277773	0.0286128	0.0106865	px^-1
ring_density_r60	normalised	0.0356582	0.0343169	0.00943703	px^-1
roots_per_100px_perimeter	normalised	2.6944	2.8699	1.3388	per 100 px
ring_attached_length_r30_px	network v2	193.667	198.5	115.79	px
ring_attached_length_r60_px	network v2	582.667	607	335.94	px
ring_length_w_r30_px	network v2	275.104	269.415	124.542	px
ring_length_w_r60_px	network v2	904.009	889.423	284.908	px
sholl_crossings_r10	network v2	6.4167	6	3.8248	count
sholl_crossings_r20	network v2	9.5	9	4.338	count
sholl_crossings_r30	network v2	11.4167	11	3.8248	count
ring_width_mean_r30_px	width	6.2875	6.2181	0.4432	px

Per field
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measure	value	units
analysed_area_px2	1.02424e+06	px^2
total_skeleton_length_px	35034	px
canalicular_length_density_per_px	0.0342048	px^-1
junction_count	776	count
junction_density_per_px2	0.00075763	px^-2
lacuna_count	12	count
lacunae_per_px2	1.172e-05	px^-2
skeleton_component_count	196	count

mean_component_length_px	178.745	px
median_component_length_px	19.5	px
field_length_density_w_per_px	0.0389428	px^-1
field_density_without_flagged_per_px	0.0342048	px^-1
field_density_in_roi_per_px		px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	7.0396	px
width_median_px	7.2111	px
width_p10_px	4	px
width_p90_px	10	px
width_pixels_used	35034	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2494	90.6112	36.7713	2.4642	0.914	0.9096	0.4535	88.1403	271.181	False
2	3004	86.0791	52.8366	1.6292	0.7895	0.772	0.3181	207.791	919.033	False
3	1698	63.4601	34.3535	1.8473	0.8408	0.955	0.5365	257.425	340.805	False
4	1631	87.5017	30.956	2.8266	0.9353	0.741	0.0248	297.032	38.7695	False
5	1204	55.3793	28.6478	1.9331	0.8558	0.9177	0.1008	394.263	364.613	False
6	2066	81.299	34.8468	2.333	0.9035	0.8619	0.1545	472.247	745.532	False
7	1233	57.0332	27.8479	2.048	0.8727	0.9278	0.052	459.387	484.83	False
8	2066	100.448	26.9826	3.7227	0.9632	0.9018	0.236	550.262	877.61	False
9	2656	108.838	32.5449	3.3442	0.9542	0.8877	-0.0333	642.811	193.327	False
10	2113	91.9731	30.2288	3.0426	0.9444	0.9135	0.0369	728.295	633.991	False
11	2173	86.8531	32.2149	2.6961	0.9287	0.9358	0.0768	787.166	480.067	False
12	1997	68.5432	38.3841	1.7857	0.8285	0.9402	0.1037	859.482	288.042	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	8	367	1177	3847.87	164	23.4626	6.9587
2	11	457	1251	2524.47	117	21.5767	5.985
3	9	268	843	2315.25	94	24.6303	6.2014
4	8	316	643	1132.45	69	16.4123	5.4451
5	4	168	539	955.607	45	21.2357	6.1518
6	8	302	926	4481.16	181	24.7578	6.5064
7	1	87	597	684.5	22	31.1136	6.5916
8	1	115	503	170.297	5	34.0593	6.1981
9	5	162	510	2778.71	122	22.7763	6.4019
10	6	214	810	4436.69	178	24.9252	6.9788
11	7	315	962	5019.43	223	22.5086	6.2349
12	4	182	749	1994.4	91	21.9165	5.7961

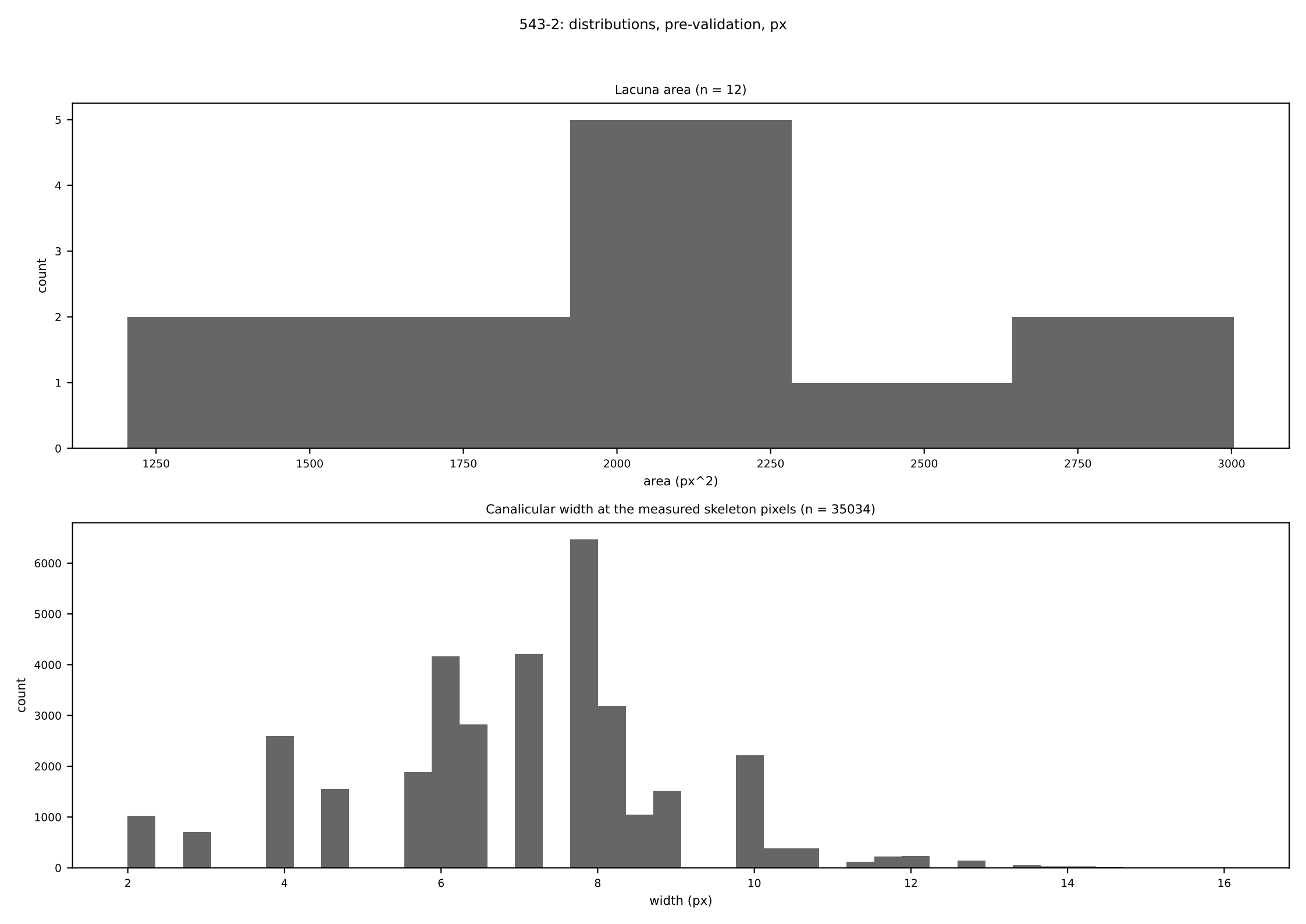
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter

1	244.492	9510	23791	1	0.9843	0.038591	0.0494725	3.2721
2	295.12	10519	26271	1	1	0.0434452	0.0476191	3.7273
3	172.894	7548	18977	1	1	0.0355061	0.0444222	5.2055
4	232.551	8756	18925	0.9575	0.7928	0.0360895	0.0339762	3.4401
5	154.267	7010	17818	1	1	0.0239658	0.0302503	2.5929
6	231.764	9080	22988	1	1	0.0332599	0.0402819	3.4518
7	155.924	7017	19168	1	1	0.0123985	0.0311457	0.6413
8	254.25	9429	23720	1	1	0.0121964	0.0212057	0.3933
9	277.078	10094	25807	1	1	0.0160491	0.0197621	1.8045
10	227.622	8939	23523	1	1	0.02394	0.0344344	2.6359
11	225.522	8872	23390	1	1	0.035505	0.0411287	3.1039
12	193.723	8132	21901	1	1	0.0223807	0.0341994	2.0648

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	321	981	419.09	1354.28	9	16	16
2	390	1139	509.233	1408.13	14	18	17
3	207	667	299.25	975.791	9	10	14
4	289	547	352.421	716.671	10	11	8
5	101	337	182.125	626.47	4	6	8
6	288	831	344.149	1058.82	8	11	10
7	24	98	95.598	677.428	1	5	7
8	29	83	132.297	584.529	1	5	9
9	146	385	178.539	579.245	5	7	7
10	190	683	239.581	918.181	5	5	12
11	227	851	351.593	1087.89	7	12	17
12	112	390	197.368	860.666	4	8	12


543-2: distributions, pre-validation, px
Lacuna area (n = 12)
count
area (px^2)
0
1
2
3
4
5
1250
1500
1750
2000
2250
2500
2750
3000
Canalicular width at the measured skeleton pixels (n = 35034)
count
width (px)
0
1000
2000
3000
4000
5000
6000
2
4
6
8
10
12
14
16